Global Lithium Carbonate Market

Analysis and Five-Year Outlook 2025–2030

A Quantitative Researcher's Perspective on Supply, Demand, and Trading Implications

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April 2026

Research, analysis, and intellectual framework by the author. Writing assistance, formatting, and editorial support provided by Claude (Anthropic AI).

Executive Summary

Lithium carbonate is at a structural inflection point. After the most severe price correction in its modern history — an 85% decline from the 2022 peak — the market has transitioned from deep surplus to the early stages of a supply deficit. The recovery is driven by the collision of a mining investment drought with accelerating demand from electric vehicles, grid-scale energy storage, and AI data center infrastructure. As of April 2026, GFEX lithium carbonate futures trade at approximately 175,000 yuan per tonne (~USD 24,000/t) — already at the upper end of our base case 2027–2028 range, suggesting that our bull case timeline is materializing faster than projected.

Central Thesis

The price recovery now underway is real and structurally supported, but bounded by sodium-ion substitution at approximately USD 18,000–20,000 per tonne — preventing a repeat of the 2022 supercycle. The next three to four years represent a genuine trading opportunity within a defined corridor of USD 15,000–25,000 per tonne, with specific observable catalysts that allow disciplined position management.

Five Key Findings

1. The deficit is forming now, not later. Fastmarkets confirms the market transitioned from a 154,000-tonne surplus in 2024 to a small surplus of 10,000 tonnes in 2025, swinging to a 1,500-tonne deficit in 2026. Current market prices at ~USD 24,000/t validate the Fastmarkets view over Wood Mackenzie's more bearish surplus projection.

2. The mining investment drought guarantees medium-term tightness. Approximately 30–40% of the expected 2027–2029 supply pipeline has been deferred or cancelled during 2023–2025. Given the 5–10 year mine construction timeline, this missing supply cannot be recovered in time to prevent the structural deficit window of 2026–2029.

3. The sodium-ion ceiling fundamentally changes this cycle. Unlike 2022, sodium-ion batteries are now commercially deployed at scale. CATL's April 2026 60 GWh sodium-ion order confirms substitution is real and active. The realistic recovery corridor is USD 15,000–25,000/t, not USD 50,000–80,000/t.

4. China's dual-star strategy means it wins regardless of chemistry outcome. China controls approximately 70% of global lithium refining (IEA), 70% of battery manufacturing (BNEF), and 48% of carbonate consumption (USGS) — while simultaneously dominating sodium-ion development. Western supply chain diversification creates a persistent geopolitical premium for non-Chinese-origin material.

5. The GFEX contract creates specific trading opportunities unavailable before 2023. Calendar spreads, basis trades versus physical, GFEX–SGX integration arbitrage, and a structurally asymmetric options market all represent real opportunities while the market remains in its early efficiency phase.

Price Outlook

Year	Bear Case	Base Case	Bull Case
2025 (actual)	$9,000–11,000	$11,000–14,000	$14,000–18,000
2026	$11,000–15,000	$14,000–19,000	$18,000–24,000
2027	$13,000–16,000	$17,000–22,000	$22,000–28,000
2028	$14,000–17,000	$19,000–24,000	$24,000–30,000
2029	$13,000–15,000	$17,000–21,000	$20,000–26,000
2030	$11,000–14,000	$15,000–19,000	$18,000–24,000

Note: Current market price (April 2026) at approximately $24,000/t is already at the upper bound of the base case 2027–2028 range, suggesting the bull case scenario is materializing faster than the base case timeline projected.

Methodological Position

This report deliberately differs from standard sell-side commodity research in one important respect: it explains its forecasting methodology rather than concealing it. The price scenarios above are not predictions but conditional analytical views — each defined by specific observable assumptions, with explicit decision rules for revision as evidence accumulates. We assign rough probability weights but acknowledge they represent informed analytical judgment rather than statistically derived estimates, given that the key variables driving the outlook are characterized by Knightian uncertainty rather than quantifiable risk.

The analytical framework — supply-demand balance anchored by cost curve boundaries, supplemented by Bayesian updating logic for incorporating new evidence — is more honest than the false precision of point forecasts and more useful than the vague directional commentary common in market reports.

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Chapter 1

Market Overview — Lithium Carbonate at an Inflection Point

1.1 From Industrial Mineral to Financial Instrument

Lithium is not a new discovery. Commercial production of lithium compounds dates back to the early twentieth century, and for most of that history lithium was what industry analysts would call a quiet specialty chemical — useful, but unglamorous. Its primary applications were in ceramics and glass manufacturing, where lithium carbonate lowered melting points and improved thermal resistance; in lubricating greases that performed reliably under extreme temperatures in aerospace and military equipment; and in pharmaceuticals, where lithium carbonate remains a frontline treatment for bipolar disorder to this day. These were stable, slow-growing markets. Nobody speculated on lithium prices. Nobody needed to.

The commercialization of lithium-ion batteries by Sony in 1991 began to change this gradually, first through consumer electronics — laptops, mobile phones — then more dramatically through electric vehicles. But it was China's aggressive state-backed push into EV manufacturing after 2015, combined with the post-COVID global EV acceleration, that transformed lithium from a specialty chemical into what it is today: a strategic critical mineral at the center of the global energy transition, a geopolitical battleground between major powers, and since July 2023, a fully listed futures commodity traded on the Guangzhou Futures Exchange (GFEX).

That last development matters more than it might initially appear. Before GFEX launched its lithium carbonate futures contract in July 2023, the world's largest lithium market — China alone consumes approximately 48% of global supply (USGS, 2024) — had no standardized exchange-traded instrument. Prices were determined through bilateral over-the-counter negotiations between producers, processors, and battery manufacturers. For outsiders, pricing was opaque. For market participants, hedging price risk was difficult and costly. For global commodity traders, financial exposure to lithium was structurally limited.

The GFEX listing in July 2023, followed shortly by the Singapore Exchange (SGX) lithium carbonate contract for international participants, marked the moment lithium carbonate crossed from opaque industrial commodity into transparent, hedgeable, financially accessible market. It is not a coincidence that this transition happened at the peak of the most dramatic price cycle in the metal's history.

1.2 What Is Lithium Carbonate — and Why Not Lithium Hydroxide?

Lithium carbonate (Li_2CO_3) is the primary refined product of lithium extraction. It is not the raw ore — it is what you get after mining and processing either hard rock spodumene or brine from salt flats. The value chain flows from raw resource (spodumene ore or lithium brine), to lithium carbonate as the primary refined product (battery-grade requires minimum 99.5% purity, traded on GFEX with ticker LC), to lithium hydroxide which is further processed from carbonate for higher energy density applications.

This distinction is not merely technical — it is commercially, analytically, and financially significant. **Lithium carbonate** is the primary input for **lithium iron phosphate (LFP) batteries**. LFP batteries are cheaper, thermally stable, safer, and have longer cycle lives — but carry lower energy density per kilogram. They dominate China's EV and grid storage markets. BYD's entire passenger vehicle lineup runs on LFP. CATL supplies LFP cells to domestic Chinese automakers and increasingly to international brands including Tesla's standard range vehicles.

Lithium hydroxide is the primary input for high-nickel NMC and NCA batteries — higher energy density, preferred for premium long-range EVs. Tesla's long-range vehicles, BMW, Mercedes-Benz, and

Volkswagen's premium EVs all historically used NMC chemistry. Korean battery manufacturers — LG Energy Solution, Samsung SDI, and SK On — built their global businesses primarily around NMC.

Why is the chemistry mix shifting toward LFP — and why does this matter? Western automakers entering the EV market in 2015–2020 prioritized range as their primary differentiator, driven by consumer range anxiety and long highway distances. NMC chemistry, requiring hydroxide, delivered the energy density they needed. But the competitive boundary is now shifting in carbonate's favor: CATL's latest LFP cells achieve over 400km range, and Tesla began sourcing LFP from CATL for its standard range Model 3 and Model Y from 2021 — a landmark signal. The structural shift from NMC toward LFP in mass-market EVs is a sustained tailwind for lithium carbonate demand.

Trading connection: The LFP/NMC chemistry split creates distinct trading dynamics on GFEX. When NMC demand rises relative to LFP, physical carbonate demand from cathode manufacturers softens — but carbonate demand from hydroxide converters (who use carbonate as feedstock) remains firm. This split signal affects the basis between GFEX futures and physical spot prices, and changes the hedging behavior of different buyer categories.

1.3 Market Size, Structure, and the Global Power Map

The global lithium market was valued at approximately USD 32–37 billion in 2024–2025, with lithium carbonate representing the largest single segment by volume at approximately 52% of total refined lithium output. Global lithium carbonate consumption reached approximately 1.18 million tonnes LCE in 2024, growing toward 1.40 million tonnes in 2025 (USGS Mineral Commodity Summaries 2025, 2026).

Comparison with other major metals:

Metal	Approx. Market Size	Annual Growth	Comment
Iron ore	~$250 billion	~2–3%	Mature, slow growth
Copper	~$200 billion	~3–4%	Critical, deep liquid market
Gold	~$250 billion	~1–2%	Store of value
Nickel	~$35 billion	~4–5%	Comparable size to lithium today
Lithium	**~$32–37 billion**	**~14–18% CAGR**	**Fastest growing major metal**
Cobalt	~$8 billion	~8–10%	Smaller, also EV-driven

Lithium today is roughly the size of nickel, but growing at four to five times the rate of any comparable industrial metal. At an 18% CAGR, the lithium market approaches copper-scale within a decade. No other major industrial metal has this growth trajectory. This is precisely why GFEX launched when it did, and why global commodity traders are paying attention.

Lithium has no meaningful recycling buffer yet. Copper, aluminum, and iron all have mature secondary supply streams that buffer demand shocks. Recycled lithium accounts for less than 5% of total supply (IEA Critical Minerals Report, 2024). This means demand growth translates almost entirely into new mining demand — amplifying both upside and downside of price cycles in ways that more mature metals do not experience.

The global power map:

Upstream (mining): Australia is the world's largest lithium producer at approximately 37% of global output, followed by Chile at 20% and China at 17% (USGS, 2024). China has been systematically acquiring stakes in Australian and African mines through joint ventures, giving it upstream influence well beyond its domestic production share.

Midstream (refining — the critical chokepoint): China controls approximately 65–70% of global lithium refining and chemical processing capacity (IEA, 2024; S&P; Global Market Intelligence). This is the most important structural fact in the entire lithium market. Raw spodumene mined in Australia by Western companies — Pilbara Minerals, Albemarle — is shipped to China for conversion into battery-grade lithium carbonate. Even lithium that Western governments consider part of their 'domestic' supply chain passes through Chinese processing infrastructure.

Downstream (battery manufacturing): China produces approximately 70–80% of global battery cells (BloombergNEF 2025), led by CATL and BYD. South Korean manufacturers — LG Energy Solution, Samsung SDI, SK On — hold significant but declining share. Western manufacturers remain a small fraction despite recent investment.

End markets: China consumes approximately 48% of global lithium carbonate output (USGS, 2024). Europe is the second largest demand region. The United States is significant but politically volatile. The Middle East is emerging as a new demand center driven by grid storage rather than EVs.

How the world is responding to Chinese dominance:

The European Union has been the most strategically coherent respondent. Through REPowerEU and the Critical Raw Materials Act (enacted 2024), the EU has committed to processing at least 40% of its lithium domestically by 2030, established direct offtake partnerships with Australian and Chilean producers, and is funding domestic gigafactory construction at scale. The EU Battery Regulation 2023/1542 mandates minimum 80% lithium recycling efficiency by 2031. The EU's motivation is acute — it has essentially no domestic lithium production, the largest non-Chinese EV market, and the recent traumatic experience of Russian energy dependency that has made supply chain security a political priority. **The EU, not the United States, represents the most credible Western challenge to Chinese midstream dominance.**

The United States has committed $3 billion through the Bipartisan Infrastructure Law for domestic lithium extraction, processing, and recycling (US DOE, 2023), but faces structural headwinds. The political inconsistency between the Biden-era IRA incentives and the Trump administration's skepticism of EV mandates has created investment uncertainty.

Japan and South Korea have pursued long-term bilateral offtake agreements with Australian and Chilean miners that partially bypass Chinese refining. Australia is attempting to move up the value chain by building domestic lithium hydroxide refining. Chile and Argentina face nationalization tensions — Chile's 2025 Codelco-SQM joint venture being the most significant example.

The net result is a market bifurcating into two partially separate supply chains — a Chinese-integrated chain that is cost-efficient and scale-dominant, and an emerging Western-aligned chain that is more expensive but commands a geopolitical premium. For a global commodity trader positioned to operate across both chains, this bifurcation creates structural trading opportunities that will define the market for years.

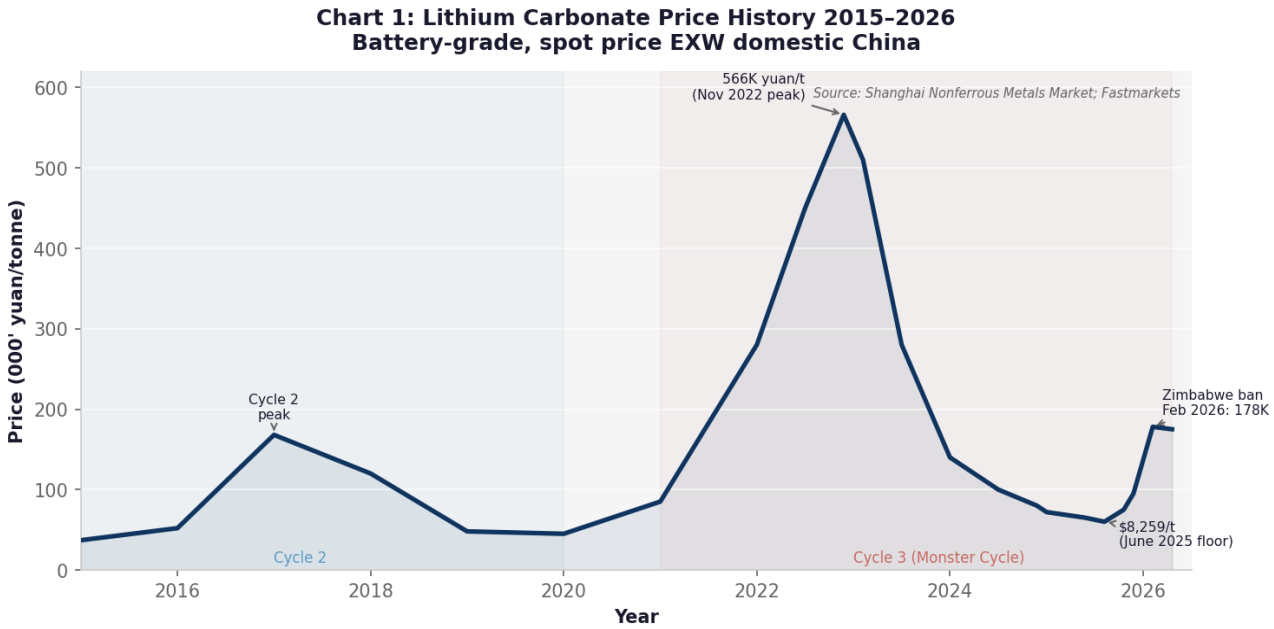
1.4 The Price Cycle — Understanding the Paradox

At first glance, the lithium carbonate market presents a paradox that demands explanation. Global EV sales reached approximately 20.7 million units in 2025 (IEA Global EV Outlook 2025), up from 14 million in 2023. Grid-scale battery storage deployments are accelerating. AI data centers are creating entirely new categories of battery demand. And yet lithium carbonate prices in mid-2025 sat at approximately $8,259 per tonne — their lowest level since 2021, and over 85% below their 2022 peak (Fastmarkets, 2025). **How does the world's most strategically critical battery material get cheaper precisely when demand for it is growing fastest?**

The answer lies in a structural characteristic shared by most hard-rock commodity markets but manifesting with particular severity in lithium: **the supply response lag**. Building a lithium mine from discovery to commercial production requires five to ten years. The market structurally cannot respond to demand signals in real time, and overshoots in both directions.

Lithium has experienced three distinct cycles. The Electronics Cycle (late 1990s–early 2000s) was gradual. The First EV Cycle (2015–2020) saw prices triple between 2015 and 2017, then crash through 2019–2020 as supply caught up. The Monster Cycle (2021–present) was the most extreme: prices rose from approximately 52,000 yuan per tonne in early 2021 to a peak of 566,000 yuan per tonne in late 2022 — a tenfold increase in under two years (Shanghai Nonferrous Metals Market). Global lithium production then jumped from approximately 737,000 tonnes LCE in 2022 to nearly 1.2 million tonnes by 2024 — a 60% supply increase in two years (Fastmarkets, 2025; USGS, 2025) — arriving precisely as EV demand growth decelerated. The resulting market surplus drove prices down by over 85% from peak.

The market bottomed in approximately June 2025 at $8,259 per tonne (Fastmarkets, 2025). By December 2025, prices had recovered toward $13,000/t. As of April 2026, GFEX lithium carbonate futures trade at approximately 175,000 yuan per tonne (~USD 24,000/t) following Zimbabwe's February 2026 export ban which removed approximately 7% of global spodumene supply, combined with strong demand signals and Ganfeng chairman Li Liangbin's projection of 30–40% demand growth in 2026.



Chapter 2

Supply Analysis — The Anatomy of a Constrained Market

2.1 Two Types of Lithium Resources — Brine vs Hard Rock

Global lithium supply comes from two fundamentally different geological sources. Understanding the distinction matters commercially because the two source types have different cost structures, production timelines, and supply response speeds that directly affect how the market behaves under price pressure.

Brine deposits are underground reservoirs of lithium-rich saltwater found primarily beneath the salt flats of South America's Lithium Triangle. Extraction involves pumping brine to the surface and allowing it to evaporate in large open-air ponds over 12 to 18 months before chemical processing. Chilean brine operations produce lithium carbonate at among the lowest costs globally, estimated at USD 4,000–6,000 per tonne (Benchmark Mineral Intelligence, 2024). The disadvantages are the long evaporation timeline and high water consumption.

Hard rock deposits — primarily spodumene pegmatite ore — are found principally in Australia, with significant deposits also in China, Zimbabwe, and increasingly Africa. Hard rock production is faster to ramp (3–5 years versus 5–8 years for new brine) but carries higher operating costs of USD 6,000–9,000 per tonne of carbonate equivalent for Australian operations (Wood Mackenzie, 2024).

The cost structure difference creates a **natural production hierarchy** determining who cuts output first when prices fall. At the June 2025 price floor of $8,259/t, essentially all hard rock operations globally were producing at or below cash cost, driving the wave of mine closures that is now tightening supply.

Trading connection: When spot prices approach the hard rock cash cost threshold of approximately $8,000–9,000/t, experienced traders recognize this as a structural price floor — sustained production below cash cost is self-correcting. Tracking the cost curve is fundamental to commodity supply analysis.

2.2 Where the Lithium Is — Global Resource Geography

Before mapping global geography, a definitional point. **Resources** refer to all lithium estimated to exist; **reserves** refer to the portion economically extractable today. According to USGS Mineral Commodity Summaries 2026, global lithium resources total approximately 150 million tonnes (revised up from 105 million previously), while proven economic reserves stand at approximately 30 million tonnes.

At projected 2025 consumption, proven reserves represent 50+ years of supply; total resources extend beyond 150 years. **Lithium is not running out.** The supply problem is not physical scarcity — it is extraction speed, processing infrastructure, and geopolitical access.

The Lithium Triangle (Chile, Argentina, Bolivia) holds approximately 56% of global lithium resources. Chile's National Lithium Strategy (2023) culminated in a 2025 Codelco-SQM joint venture nationalizing 50% of Chile's primary producer. Argentina's federal structure allows more commercial flexibility — output doubled in 2024. Bolivia holds the largest reserves but remains absent from supply due to extraction challenges.

Australia produces approximately 88,000 tonnes of lithium per year — 37% of global output (USGS, 2024) — primarily from hard rock spodumene including the world's largest mine at Greenbushes. Australia's structural challenge is exporting most production as raw concentrate to China for processing.

Zimbabwe produced approximately 22,000 tonnes in 2024 — 9% of global output — driven almost entirely by Chinese investment. The country's **February 2026 ban on lithium concentrate exports** removed approximately 7% of global spodumene supply overnight and was a major catalyst for the price spike to

175,000 yuan/t in early 2026.

2.3 The Midstream Chokepoint — China's Refining Dominance

China controls approximately 65–70% of global lithium refining and chemical processing capacity (IEA, 2024). This is the most important structural fact in the entire market. The global lithium supply chain, regardless of where ore is mined, flows through Chinese processing infrastructure.

The hidden sulfur vulnerability: Sulfuric acid is critical for converting spodumene to lithium carbonate, and China imports significant sulfur from the Middle East. The closure of the Strait of Hormuz during the 2025–2026 US-Iran conflict therefore affected lithium not only through shipping disruptions but through tightening of refining inputs (Energy Storage News, 2025).

Trading connection — refinery operating rates as signal: Chinese refinery operating rates are a critical leading indicator for physical carbonate availability. When refinery margins compress (carbonate price falls below conversion cost), Chinese processors cut production, tightening physical supply. The spodumene-to-carbonate spread — calculated from Pilbara BMX prices versus GFEX carbonate futures — functions as lithium's version of the oil refiner's crack spread. Weekly inventory data from Shanghai Metals Market and Mysteel provides a near-term supply signal that directly influences GFEX futures positioning.

The two-tier physical market and geopolitical premium: Western buyers seeking IRA compliance or EU battery regulation compliance increasingly need to demonstrate non-Chinese-processed lithium. This has created a price premium for Western-origin carbonate. For a global trader with both Chinese and Western supply chain access, the ability to source, certify, and deliver Western-origin material to premium buyers is a structural commercial advantage.

2.4 Key Players — Who Controls Supply

Western Upstream:

Albemarle (NYSE: ALB) — largest Western lithium producer, ~17% global market share. Operations in Chile (Salar de Atacama with SQM), Australia (49% Greenbushes), Nevada (Silver Peak). Implementing significant production cuts during the downturn.

SQM (NYSE: SQM) — Chilean producer with lowest-cost brine assets at Salar de Atacama. Pursuing 'volumes-over-value' strategy targeting 240,000 tonnes carbonate capacity by 2026. Codelco took 50% via JV in 2025.

Pilbara Minerals (ASX: PLS) — Australia's largest pure-play lithium miner. P1000 expansion completed January 2025. The BMX auction platform is watched as a global price discovery mechanism.

Chinese Integrated Players:

Ganfeng Lithium (SZSE: 002460) — China's most globally aggressive lithium company. Supply deals with Tesla, BMW, LG Chem, Volkswagen, Hyundai. Goulamina (Mali) operational December 2024. August 2025: $1.8B Argentina JV targeting 150,000 tonnes/year LCE. Chairman Li Liangbin's February 2026 projection of 30–40% demand growth in 2026 contributed to the recent price spike.

Tianqi Lithium (SZSE: 002466) — controls 51% of Greenbushes through Talison alongside Albemarle, giving China effective control over the world's largest lithium mine.

The Battery Maker That Moves Everything:

CATL (SZSE: 300750) — world's largest battery manufacturer. CATL's decisions on chemistry, procurement, and upstream investment make it the single most market-moving entity in global lithium. CATL shut its Jiangxi lepidolite mine in August 2025 (~3% of global supply), and signed the world's largest sodium-ion battery order (60 GWh with HyperStrong) in April 2026 (Caixin Global, 2026).

Rio Tinto acquired Arcadium Lithium for $6.7 billion in 2024, creating Rio Tinto Lithium targeting 200,000+ tonnes LCE annually by 2028 — a major diversified miner making a multi-billion dollar bet on lithium at the bottom of the price cycle.

2.5 The Mining Investment Cycle — Seeds of the Next Shortage

The most important supply-side dynamic for a 5-year price outlook is not current production levels — it is the investment decisions being made, or more precisely not being made, right now. Mine development requires 5–10 years and hundreds of millions to billions of dollars of upfront capital. The supply available in 2027–2028 is largely determined by investment decisions made in 2020–2021.

Sustained prices below the cost of production triggered investment cancellations and mine closures across the industry: Albemarle deferred multiple expansion projects; Arcadium placed Mt. Cattlin into care and maintenance; CATL shut its Jiangxi mine in August 2025; Zimbabwe's export ban froze new investment. **Fastmarkets estimates the global lithium project pipeline has been reduced by approximately 30–40% from its 2022 peak** (Fastmarkets, 2025). The projects not built today are the missing supply of 2027 and 2028.

Current consensus projections suggest the market moves from a 154,000-tonne surplus in 2024 to a small surplus of 10,000 tonnes in 2025, then a 1,500-tonne deficit in 2026 (Fastmarkets, 2025). Wood Mackenzie projects a minor surplus continuing through 2026–2027, with structural deficits emerging only after 2030 — a divergence that represents the single most analytically important uncertainty in this report. **Current market evidence (April 2026 prices ~$24,000/t) appears more consistent with the Fastmarkets view.**

2.6 Supply Wildcards — Direct Lithium Extraction and Recycling

Direct Lithium Extraction (DLE) extracts lithium from brine in hours or days rather than 12–18 months. If commercial at scale, it would fundamentally change supply response dynamics. Current DLE pilot production costs run approximately $8,000–12,000/t — competitive with high-cost hard rock. **The estimated activation threshold is sustained prices above approximately $15,000–18,000/t**, above which DLE projects become bankable. DLE projects with 2–3 year development timelines (vs 5–10 for conventional) could bring supply faster than any previous technology — providing an additional structural cap on price duration.

Companies closest to commercial scale: **EnergySource Minerals** (Salton Sea geothermal, targeting 20,000 tonnes/year by 2026); **Standard Lithium** (Arkansas oil field brine with Koch Industries); **Lilac Solutions** (licensed to Rio Tinto's Rincon project Argentina); **International Battery Metals** (modular DLE units).

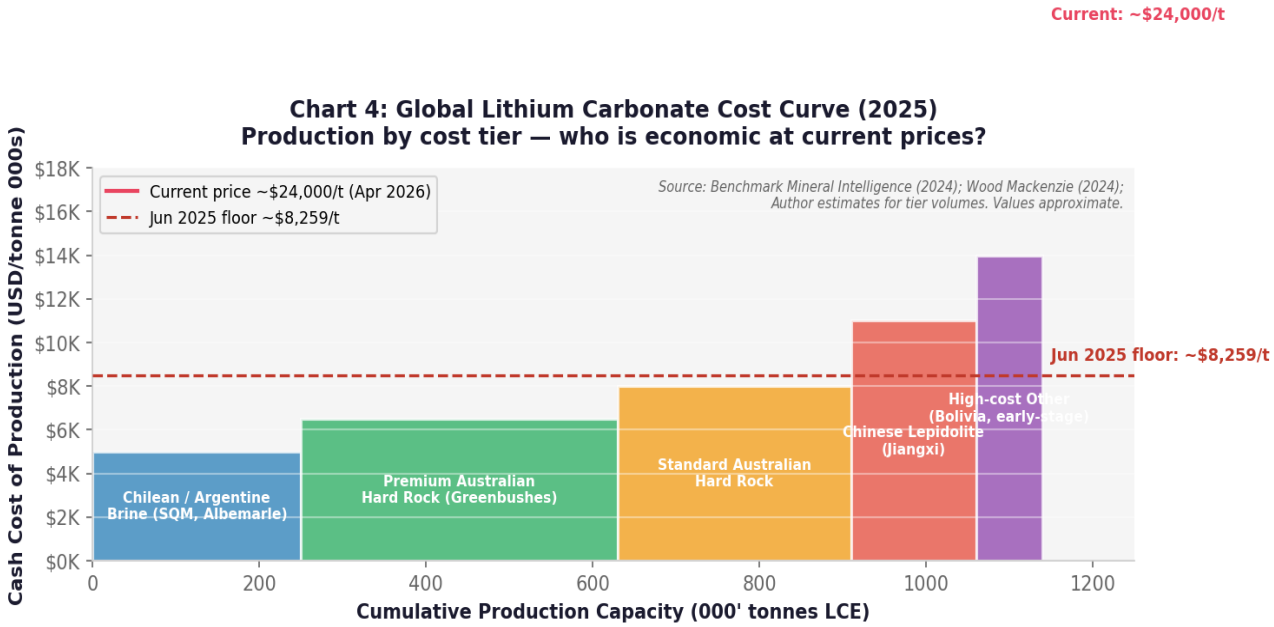
Lithium recycling currently accounts for less than 5% of total supply (IEA, 2024). By 2030, BloombergNEF projects recycled lithium could supply 8–12% of global demand. Beyond 2030, recycling becomes structurally significant. For the 5-year horizon of this report, recycling is an emerging dynamic to monitor rather than a market-moving factor.

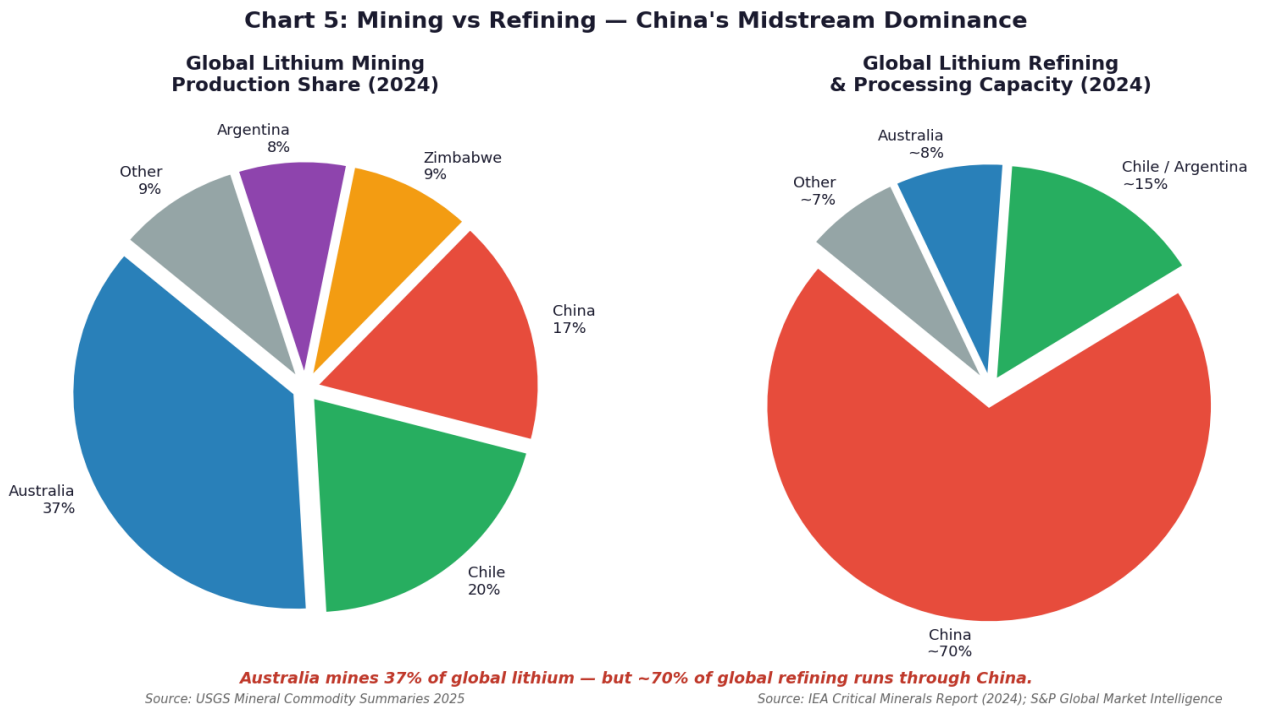
2.7 Supply Outlook — Putting It Together

Near-term (2025–2026): Supply tightening from mine closures, project deferrals, and the Zimbabwe export ban. Market transitioning from surplus to small deficit.

Medium-term (2027–2028): Period of greatest supply risk. Investment drought of 2023–2025 means new project supply will be limited. Demand continues growing. Deficit projected to widen unless idled capacity is restarted at scale or DLE projects advance faster than expected.

Longer-term (2029–2030): New investment prompted by 2026–2027 price recovery begins to arrive partially. Recycling contributes growing share. Market begins to rebalance — though whether this produces another oversupply or sustainable equilibrium depends on industry investment discipline.





Chapter 3

Demand Analysis — Growth, Complexity, and the Questions Worth Asking

3.1 The Demand Framework — Three Categories That Behave Differently

Lithium carbonate demand comes from three structurally different categories. **Category 1 — Existing demand** (EVs in China, consumer electronics, industrial) — established, growth predictable. **Category 2 — Scaling demand** (grid storage, commercial EVs) — happening now, accelerating, policy-sensitive. **Category 3 — Emerging demand** (AI data centers, humanoid robots, electric aviation) — early stage, high uncertainty, could be transformative or could disappoint. Most lithium demand forecasts mix these together into a single growth number, implicitly treating emerging demand with the same confidence as established demand. The result is forecasts that look precise but are highly sensitive to unstated assumptions.

3.2 Electric Vehicles — The Dominant and Most Complex Driver

Electric vehicles remain the largest driver of lithium carbonate demand, accounting for approximately 65–70% of total battery demand in 2024 (IEA Global EV Outlook, 2025). **Global EV sales reached approximately 17 million units in 2024 and 20.7 million units in 2025** (IEA, 2025), representing more than 25% of total new light vehicle sales globally for the first time.

If EV sales nearly doubled between 2023 and 2025, why did lithium carbonate prices fall 85% over the same period? The paradox resolves when you decompose EV demand into its component parts:

First, the **chemistry split** matters enormously. Only LFP-battery EVs consume lithium carbonate directly. NMC-battery EVs consume lithium hydroxide. In China, approximately 70% of EVs sold in 2024 used LFP chemistry (GGII, 2025). Globally, LFP's share has risen from ~30% in 2020 to ~55% in 2024 (BNEF, 2025).

Second, **lithium intensity per vehicle is falling** — approximately 3–5% per year through better cell design (Wood Mackenzie, 2024). A vehicle requiring 600kg of LCE in 2024 may require only 480–500kg by 2030. At scale of tens of millions of vehicles, this intensity decline is a meaningful structural headwind that headline forecasts miss.

Third, **battery pack right-sizing** is becoming commercial reality. BYD's Seagull — China's best-selling EV in 2024–2025 — has a 38kWh battery pack, less than half the size of premium long-range vehicles. As EV adoption reaches mainstream consumers prioritizing cost over range, average pack sizes may fall.

Net calculation: EV volume growth of +80% (22M to ~37M units 2024to2030) combined with LFP share increase, intensity decline, and right-sizing produces approximately +55–60% lithium carbonate demand growth from EVs — still strong but materially lower than headline EV growth implies.

Regional picture: China dominates with ~60% of global EV sales (almost entirely LFP). Europe ~20% (mixed, shifting to LFP). United States ~8–10% (predominantly NMC, structurally and politically constrained). Japan, Korea, emerging markets ~10–12%. **Global lithium carbonate demand from EVs is determined primarily by China, secondarily by Europe, and only marginally by the United States.**

3.3 China's Domestic Policy — The Single Most Market-Moving Variable

If there is one section of this report that a lithium carbonate trader must internalize above all others, it is this one. **A 10% change in Chinese EV policy has more impact on global lithium carbonate demand than the entire US EV market combined.**

China's EV demand is not primarily market-driven — it is policy-engineered. The NEV mandate requires automakers to meet minimum NEV sales quotas. Vehicle purchase tax exemptions for NEVs are worth 10,000–30,000 yuan per vehicle. National battery standards progressively favor LFP through safety and subsidy eligibility criteria. Trade-in subsidy programs in 2024–2025 directly stimulated demand during the slowdown.

Sensitivity analysis: China EV growth at 20%/year (bull) implies ~680,000t carbonate demand from China EVs alone in 2028; at 15% (base) ~515,000t; at 8% (bear) ~356,000t. **The difference between bull and bear is ~324,000 tonnes — roughly 35% of total 2024 global production.** No other single variable has comparable impact.

Risks to Chinese EV demand: the property sector overhang remains the most significant near-term risk; market saturation in Tier 1/2 cities (already >50% EV penetration); price war dynamics damaging industry profitability. **What to watch:** Monthly CAAM NEV sales data is the single most important data release for lithium carbonate demand assessment.

3.4 EU, US, and Emerging Markets

The European Union represents the most credible non-Chinese demand growth market. The 2035 ban on new ICE vehicle sales is legally binding and embedded in EU law — significantly more durable than US executive orders. European EV sales reached ~3.2 million units in 2024 (~22% market share) (ACEA, 2025). The Critical Raw Materials Act mandates 40% domestic processing by 2030.

The United States presents the starkest policy uncertainty of any major EV market. The Biden IRA created comprehensive EV demand stimulus; the Trump administration's rollback has introduced profound uncertainty. Combined with structural headwinds (geography, culture, economics), US EV demand faces real challenges. However — **the US matters less for lithium carbonate demand than its economic size suggests** because US EVs are predominantly NMC (hydroxide) chemistry.

Emerging markets represent long-term optionality. India's EV market reached ~2 million units in FY2025 but is concentrated in small vehicles. Southeast Asia (Indonesia, Thailand, Vietnam) is emerging as both manufacturing hub and growing market. The Middle East is a major new demand center driven by grid storage, not EVs.

3.5 Grid-Scale Energy Storage — The Fastest Growing Segment

Grid-scale BESS is growing faster than EVs as a share of lithium carbonate demand. Annual global deployments reached approximately 247 GWh in 2025 (BloombergNEF, October 2025), up from 155 GWh in 2024. BNEF forecasts annual deployments reaching 360 GWh in 2026 and 520–700 GWh by 2030. **This represents a 3–4x increase in annual deployments over five years.**

Grid storage applications are almost entirely LFP — making this one of the purest carbonate demand drivers with minimal hydroxide competition. The Middle East has emerged as a major BESS deployment market — CATL and BYD secured ~42 GWh of energy storage orders in the Middle East for 2025 (Caixin Global, 2026), with Saudi Arabia and UAE ranking as China's third and fifth largest overseas energy storage markets.

Could V2G technology disrupt this? Vehicle-to-grid technology could allow EV batteries to serve as distributed grid storage. The risk is real but unlikely to be market-moving within the 5-year horizon — grid operators require predictable, dispatchable storage that V2G cannot reliably provide at scale.

3.6 AI Data Centers — Structural New Demand

Worldwide data center electricity consumption is projected to rise from 448 TWh in 2025 to approximately 980 TWh by 2030 (Gartner, 2025) — more than doubling in five years. AI-optimized servers alone are projected to increase fivefold from 93 TWh to 432 TWh. **Global investment in data centers will overtake investment in crude oil supply for the first time in 2025** (IEA World Energy Outlook 2025).

Global shipments of AI data center energy storage lithium-ion batteries are projected to exceed 300 GWh by 2030, growing from approximately 12 GWh in 2025 (BloombergNEF, 2025).

Important methodological note on double-counting: Much of what is reported as 'AI data center battery demand' is more accurately grid-level storage demand driven by data center electricity consumption — already captured in BESS projections. The genuinely incremental carbonate demand from AI data centers — on-site UPS, power quality management — is more conservatively estimated at 50–100 GWh by 2030 rather than the 300 GWh headline figure (Wood Mackenzie, 2025). Analysts using the headline figure overstate incremental demand by approximately 200–250 GWh equivalent to 50,000–60,000 tonnes LCE.

3.7 Demand Risk — Substitution, Efficiency, and Structural Headwinds

This section deliberately addresses demand risks that bullish lithium reports minimize — because intellectual honesty about downside scenarios distinguishes rigorous analysis from marketing material.

Sodium-ion substitution by segment: Short-range urban EVs (<200km) — already viable, ~15–20% of EV demand at risk. Standard-range EVs (200–400km) — approaching viability with CATL Naxtra gen-2, price trigger $18,000–20,000/t, ~25–30% at risk. Long-range premium EVs — not viable, protected. Grid storage — highly viable, price trigger $12,000–14,000/t, significant portion at risk above this level.

Battery efficiency improvement (the silent headwind): Energy density improving 3–5% annually. Implication at scale: at 40 million EVs x 60kWh average pack in 2030, with intensity dropping from 8kg/kWh to 6kg/kWh, the demand reduction is approximately 3.8 million tonnes LCE — roughly 4x current global production. This is not a trivial adjustment — demand forecasts that project EV volume growth without incorporating intensity decline systematically overstate carbonate demand by 15–20%.

Demand sensitivity table: Base case 2030 demand of approximately 1.6M tonnes LCE. Bear adjustments: sodium-ion substitution -80kt, efficiency improvement -150kt, right-sizing -50kt, V2G moderating storage -40kt, economic slowdown -100kt = bear case ~1.18M tonnes. Bull adjustments: faster AI adoption +80kt, Middle East acceleration +60kt, India earlier than expected +70kt, faster grid storage +90kt = bull case ~1.90M tonnes.

3.8 A Question Worth Asking: Could Demand Fall in the Near Future?

Before accepting the demand growth narrative, it is worth examining the uncomfortable question: is there a realistic scenario where lithium carbonate demand falls in absolute terms? **The honest answer: no, demand will not fall in the near future. But the growth rate is slower and more nuanced than headline EV numbers suggest.**

Volume demand (number of batteries) keeps growing. Intensity demand (lithium per battery) is falling. Net demand grows but slower than volume alone implies. Even in the bear case, 2030 demand of ~1.18 million tonnes represents ~28% growth from 2024 levels. **The critical variable for price is not demand alone but the gap between demand and supply.** Even modest demand growth creates a price-moving deficit if supply has been cut as aggressively as it has during 2023–2025.

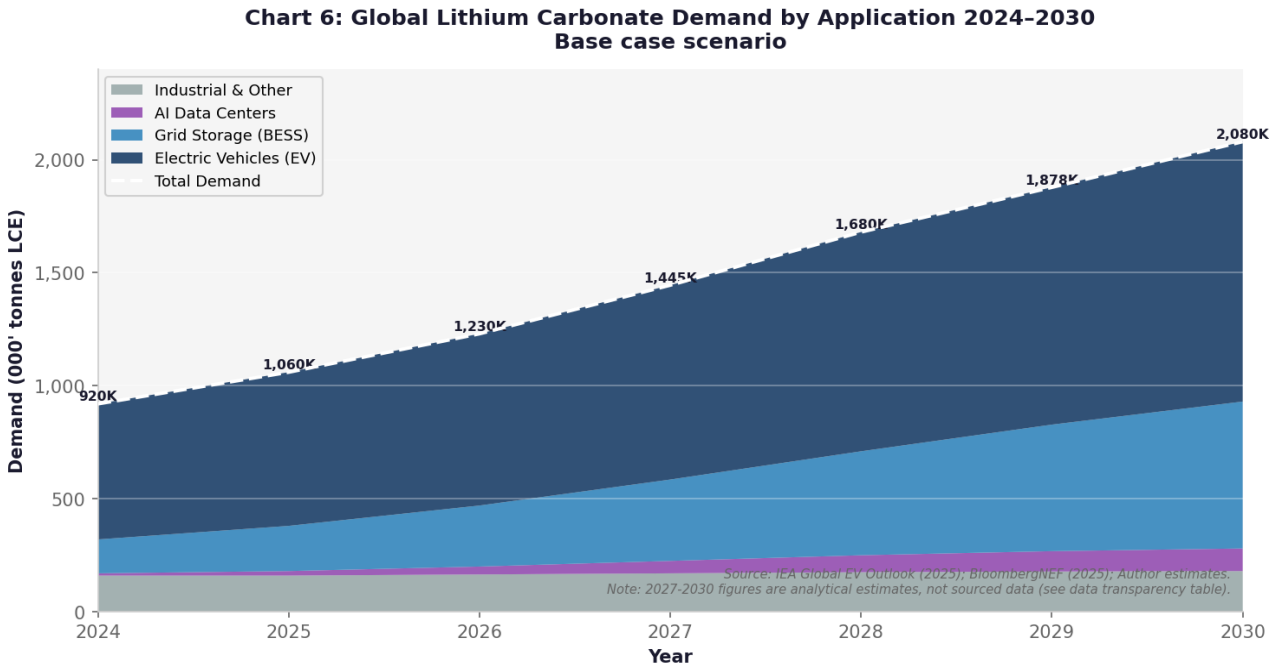
3.9 Trading Connections — Reading Demand Signals

Demand-side trading dynamics differ from supply-side because demand is harder to observe in real time. Understanding which indicators lead, which lag, and by how much is one of the most practically valuable skills in lithium trading.

Leading indicator hierarchy: (1) *Monthly CAAM NEV sales* — most important single demand signal, leads GFEX prices by 4–8 weeks; sustained acceleration above 20% YoY signals demand growth not yet priced. (2) *GGII battery installation data* — more direct read on actual carbonate consumption, distinguishes LFP from NMC. (3) *CATL and BYD quarterly earnings guidance* — particularly inventory and order book commentary, leads official data by 4–6 weeks.

Demand seasonality: Chinese EV sales spike in December (year-end push) and March-April (post-CNY ramp). Battery makers build inventory 6–8 weeks ahead. October-November contracts trade at premiums; February contracts at discounts. **The calendar spread strategy**: long October/short February captures systematic seasonal premium with low directional exposure.

The demand destruction threshold as price ceiling: The asymmetry between supply floor ($8,000–9,000/t) and demand ceiling ($18,000–20,000/t via sodium-ion substitution) bounds the medium-term price range. For options traders, this asymmetry has direct implications: standard symmetric volatility models systematically misprice the tail probabilities. Deep out-of-money calls above $25,000 are likely overpriced; options struck in the $14,000–20,000 recovery corridor are likely underpriced if the market has not yet fully priced the structural deficit.



Chapter 4

Price Outlook — On Forecasting, Uncertainty, and Quantitative Foundations

4.1 On the Purpose and Limits of Price Forecasting

Every serious commodity market report eventually arrives at the same moment: the price forecast. It is the section most readers turn to first, the number most frequently cited in trading decisions, and — if we are being intellectually honest — the part of the analysis most likely to be wrong.

This chapter does not begin with a price table. It begins with a more fundamental question: what is a commodity price forecast actually claiming, and how should a rigorous analyst think about the difference between what forecasting can honestly deliver and what it is conventionally presented as providing?

What forecasting claims to be: Standard commodity research presents price forecasts as outputs of analytical models. Goldman Sachs publishes '$13,250 per tonne by end-2026.' Morgan Stanley offers ranges. Bernstein projects $25,000 in 2027. These numbers are presented with institutional authority but rarely explain how they were derived.

What forecasting actually is: Commodity price forecasting is not prediction in the scientific sense. It is structured scenario construction — the systematic organization of known information into a framework that clarifies the range of plausible outcomes and the key variables that determine which outcome materializes.

Why bank reports never explain methodology: Three reasons. First, competitive advantage — methodology is the trade secret. Second, reputational protection — opacity prevents specific refutation. Third, client utility — most clients want a number to act on, not a seminar on epistemology.

The probability problem: When research reports assign 'base case 55%, bull case 25%, bear case 20%' probabilities, these numbers are not derived from formal probability models. They reflect analyst confidence dressed as statistical precision. As Frank Knight argued in 1921, there is a fundamental difference between *risk* (where outcome distribution is known) and *uncertainty* (where it is not). The question of whether China's EV growth rate will be 8% or 18% is an uncertainty question — assigning 25% probability is epistemically meaningless. Keynes made the same point: 'there is no scientific basis on which to form any calculable probability whatever.'

4.2 Our Analytical Approach — Transparency Over Precision

What we are not doing: Building a formal econometric model (data limitations make this misleading). Assigning formal probabilities to scenarios (Knightian uncertainty makes this dishonest).

What we are doing: A three-layer framework. **Layer 1 — Cost curve anchors:** hard rock cash cost floor $8,000–9,000/t and full cost incentive price $18,000–22,000/t bound the medium-term range. **Layer 2 — Supply-demand balance:** directional analysis from Chapters 2 and 3. **Layer 3 — Bayesian updating:** specifying observable conditions that revise our view as evidence accumulates.

This approach is connected to the Bayesian dynamic modeling framework applied in academic work on international market integration — where extracting a reliable signal (global market integration) from noisy observable variables (country returns) in a system subject to structural breaks is structurally analogous to extracting a reliable supply-demand signal from noisy lithium market data. The mathematical implementation differs, but the epistemological structure is identical: prior beliefs, likelihood updating, posterior revision.

4.3 Quantitative Foundations — Formalizing the Framework

> ***Important note on the values that follow:*** *The numerical values in this section are calibrated illustrative estimates intended to demonstrate the analytical structure rather than report formally estimated parameters. Rigorous estimation requires proprietary time-series data — specifically Fastmarkets' historical price assessments and project-level supply data from Wood Mackenzie — beyond the scope of this report. A formal implementation of the state-space framework described below would yield parameter estimates with associated standard errors; the values shown here represent our best analytical judgment of plausible magnitudes based on available public information.*

4.3.1 The price-balance relationship — nonlinearity and convexity:

dP(t) = a + b1 * B(t-k) + b2 * B(t-k)^2 + e(t)

where dP(t) is percentage price change in period t; B(t-k) is the supply-demand balance k periods ago (lag k approximately 2-4 months); b1 is the linear coefficient capturing direction of effect; b2 captures convexity; a is a constant; e(t) is the error term. If b2 < 0 (which historical data suggests), large deficits produce disproportionately large price increases. Calibrated approximate values: each 50,000-tonne deficit corresponds to ~15-25% price movement at small deficit levels but ~80-100%+ at deficits above 150,000 tonnes — explaining the 2022 nonlinear spike.

4.3.2 Cost curve price floor derivation:

Producers ordered by cost C(1) <= C(2) <= ... <= C(n). Market clearing: Sum_i Q_i(P) = D(P). Short-run price floor = cash cost of marginal producer. For 2025 demand ~1.06M tonnes: Tier 1 (Chilean brine, $4-6K) ~250kt, Tier 2 (premium Australian, $6-7K) ~380kt, Tier 3 (standard Australian, $7.5-9K) ~280kt, Tier 4 (Chinese lepidolite, $10-12K) ~150kt. **The June 2025 floor at $8,259/t is precisely where Tier 4 becomes uneconomic — confirming the cost curve framework.**

4.3.3 Bayesian updating — formal structure:

P(scenario | evidence) = P(evidence | scenario) x P(scenario) / P(evidence)

Illustrative example with prior P(Bull)=0.25, P(Base)=0.50, P(Bear)=0.25 and likelihood values P(CAAM>20% | Bull)=0.70, P(CAAM>20% | Base)=0.30, P(CAAM>20% | Bear)=0.05: posterior P(Bull | CAAM>20%) = 0.175 / 0.3375 = 0.519. **Three months of CAAM >20% growth would update bull probability from 25% to ~52% — more than doubling based on this single signal.**

4.3.4 State-space formulation (methodological note):

Measurement equation: y(t) = Lambda * f(t) + e(t), where y(t) is the vector of observables [GFEX price, inventory, CAAM sales, spodumene price, refinery utilization], f(t) is the latent supply-demand state, Lambda is the factor loadings matrix, and e(t) is idiosyncratic measurement error.

Transition equation: f(t) = Phi * f(t-1) + eta(t), where Phi captures persistence of supply-demand conditions and eta(t) represents structural shocks to fundamentals. This Bayesian Dynamic Factor Model formulation would allow filtering the true supply-demand signal from noisy market data, estimating regime changes, and producing probability distributions over future states. Full implementation is beyond the scope of this report — but the framework illustrates why simple regression approaches are misspecified: the latent state is not directly observable.

4.4 Data Quality and Provenance

Commodity market analysis combines data of varying quality and provenance. Unlike academic research where sources are standardized and peer-reviewed, market analysis draws on proprietary databases, public regulatory filings, company disclosures, and analytical judgment in proportions that vary by data point. The table below makes this provenance explicit — a practice standard in academic research but uncommon in market reports.

Data Category	Examples	Confidence
VERIFIED (Primary sources)	Price peak 566K yuan/t (2022); bottom $8,259/t (June 2025); surplus 175K / 154K / 10K tonnes 2023/24/25; deficit 1.5K (2026); production 737K to 1.2M LCE (2022-2024); EV sales 17M (2024) and 20.7M (2025) (IEA); China consumption 48%; batteries 87-88% end use	**HIGH**
APPROXIMATE (Industry estimates)	Hard rock cash cost $6-9K/t; brine $4-6K/t; full cost incentive $18-22K/t; China refining ~70%; sodium-ion substitution thresholds $12-20K/t; battery efficiency improvement 3-5%/yr; lithium intensity decline 8 to 6 kg/kWh by 2030	**MEDIUM**
CONSTRUCTED (Illustrative only)	Supply-demand balance projections 2027-2030; regression coefficients; Bayesian likelihood values; partial derivative magnitudes; cost curve tier volumes	**LOW (clearly labeled)**

The directional thesis — market transitioning from surplus to deficit, prices recovering from a cyclical floor, recovery structurally bounded by sodium-ion substitution — rests primarily on Category 1 verified data and is robust. The specific magnitude and timing of the deficit, and the precise price levels at which structural changes activate, depend more heavily on Category 2 and 3 estimates and should be treated as indicative rather than precise.

4.5 Why This Cycle Is Structurally Different From 2022

Structural difference 1 — Sodium-ion ceiling: In 2022, sodium-ion was a laboratory technology. Today CATL's Naxtra cells are in commercial production; the world's largest sodium-ion order (60 GWh) was signed April 2026. Sodium-ion becomes cost-competitive with LFP for grid storage at ~$12,000–14,000/t and for budget EVs at ~$18,000–20,000/t. The ceiling does not prevent recovery to $15,000–22,000/t — it prevents recovery to $50,000–80,000/t.

Structural difference 2 — GFEX hedging infrastructure: The 2022 supercycle was amplified by absence of hedging. Battery makers had no ability to lock in future prices, forcing competition for spot supply. The GFEX contract changed this dynamic structurally — distributing price risk and reducing panic amplification.

Structural difference 3 — DLE shortening supply lag: The 7–10 year mine construction lag that made 2022 so extreme is being partially shortened by DLE technology with 2–3 year development timelines. Not eliminated, but the next supply response is structurally faster than the last one.

Realistic recovery corridor: $15,000–25,000/t. The 2022 peak of ~$80,000/t was a structural anomaly enabled by absence of substitutes, hedging instruments, and short-lag supply response. That anomaly will not repeat in the same form.

4.6 Two-Cycle Framework — Locating the Current Moment

The price cycle bottomed June 2025 at ~$8,259/t — exactly at the hard rock cash cost floor. By April 2026, prices have recovered to ~$24,000/t. The price cycle is in its early recovery phase; historical recoveries take 18–30 months to reach incentive price level.

The mining investment cycle is at its trough. Investment cancellations have removed ~30–40% of expected 2027–2029 supply. Price recovery must be sustained for 12–18 months before new investment decisions are made with confidence. New supply from current decisions will not arrive before 2031–2033.

The interaction — a 3–4 year structural window. The combination creates approximately 2026–2029 during which deficit widens, prices recover toward incentive level, but new supply structurally cannot arrive. This window is not a forecast — it is a mechanical consequence of development timelines applied to current cycle position.

4.7 Price Scenarios — Conditional Views

Bear Case — 'Demand Disappointment with Supply Response':

Conditions: China EV growth 8–10%/year; sodium-ion 20%+ in grid storage by 2027; SQM volumes-over-value preventing tightening; DLE achieving early commercial scale. Price range: $10,000–17,000/t through 2028. Signal: three months of China NEV growth below 10% YoY plus CATL sodium-ion exceeding 12% before end-2026.

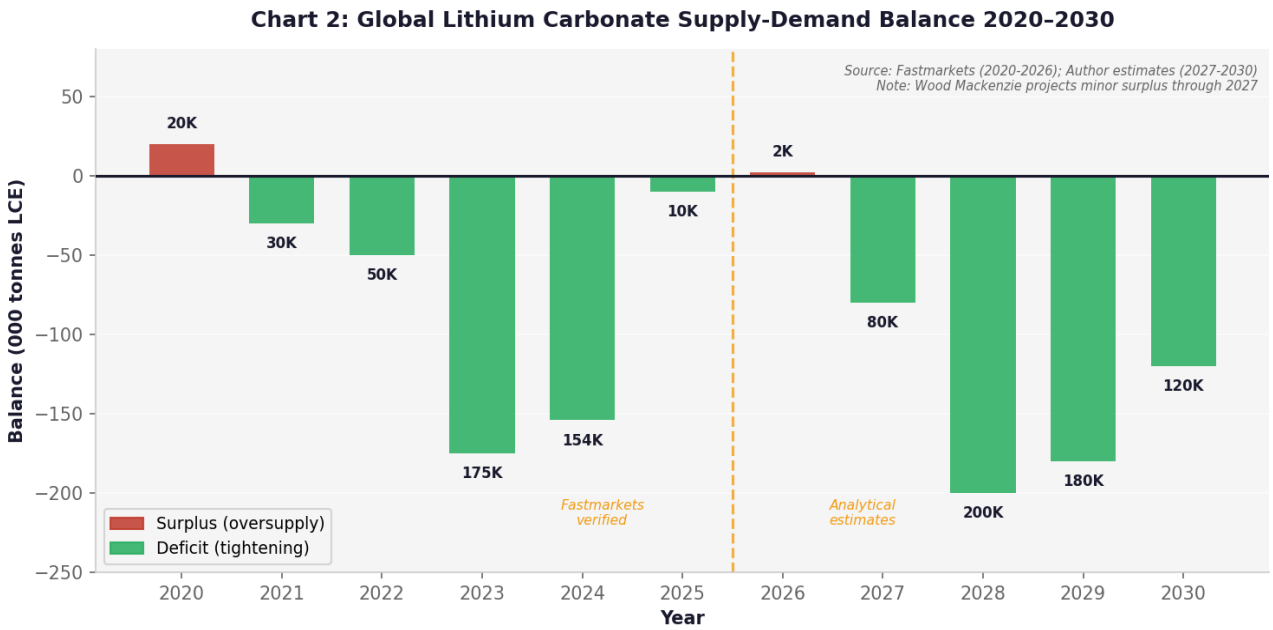
Base Case — 'Structural Deficit with Capped Recovery':

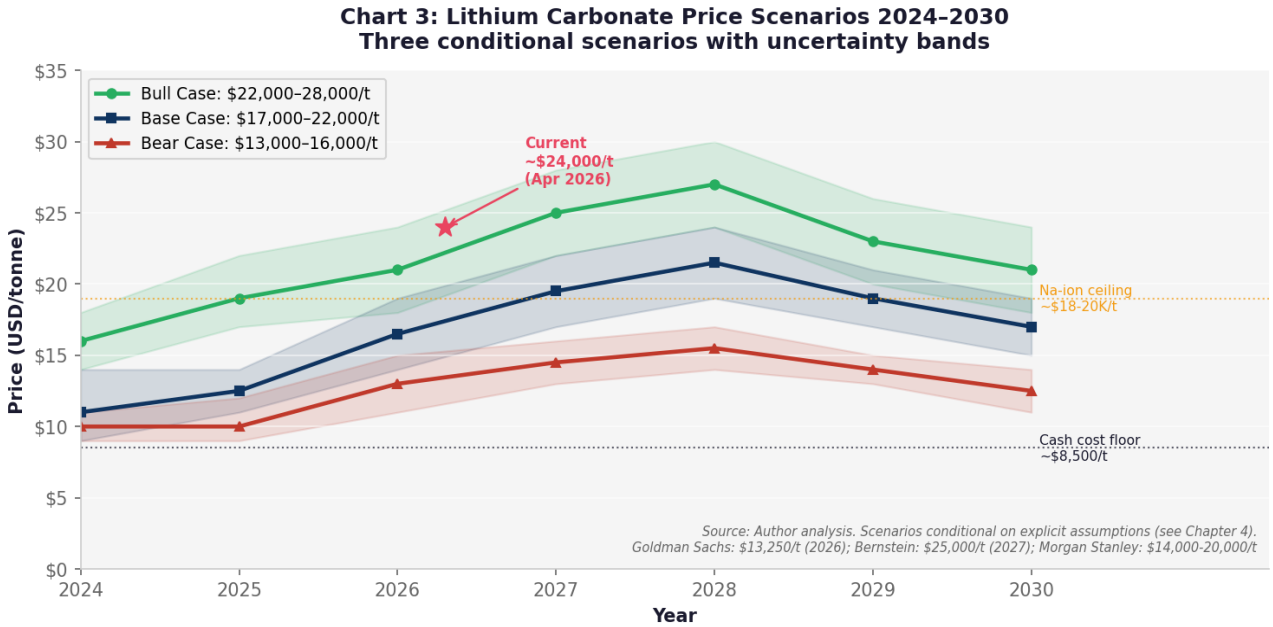
Conditions: China EV growth 12–15%/year; grid storage ~900–1,000 GWh annual deployments by 2030; mining investment drought removing 30–40% of expected 2027–2029 supply; sodium-ion modest share in budget segments. Price range: $14,000–24,000/t. Signal: China NEV sustaining 12–15% YoY through H2 2026; GFEX inventory below 40,000 tonnes; no major new mine investment above 50,000t before mid-2027.

Bull Case — 'Accelerated Demand with Extended Deficit':

Conditions: China EV growth 18–20% sustained; grid storage exceeding projections; AI data center demand arriving faster; DLE delays confirmed; additional country export restrictions. Price range: $18,000–30,000/t. Signal: three months China NEV above 20% YoY; GFEX inventory below 25,000 tonnes; additional export restrictions before end-2026.

Current market context (April 2026): Prices at ~$24,000/t are already at the upper bound of the base case 2027–2028 range and within the bull case range. The Zimbabwe export ban (February 2026) and Ganfeng chairman's projection of 30–40% demand growth in 2026 have accelerated the timeline — broadly consistent with our structural thesis but with the deficit forming earlier and more sharply than Fastmarkets' base case projected.





4.8 Comparison to Bank Consensus

Goldman Sachs (March 2026): ~$13,250/t for 2026. Morgan Stanley (Q1 2026): $14,000–20,000/t. Bernstein (2025): ~$25,000/t for 2027. Our base case: $17,000–22,000/t for 2027.

Where we differ and why: Goldman's lower estimates likely underweight the supply impact of the 2023–2025 investment drought — assuming faster supply recovery from idled capacity than restart timelines support. Bernstein's $25,000 underweights the sodium-ion ceiling: above $20,000/t, grid storage buyers have a commercially available alternative CATL is deploying. Our base case sits between, reflecting both the genuine deficit and the structural ceiling.

4.9 Synthesis — What This Analysis Can and Cannot Tell You

What we can claim with reasonable confidence: The direction of the next price move is upward — robust across assumption variations. The recovery will be structurally bounded relative to 2022. The 3–4 year structural window is a mechanical consequence of cycle positioning.

What we cannot claim: Precise timing. Meaningful probabilities for the scenarios. Guaranteed accuracy of supply-demand numbers given imperfect public data.

The most important thing this analysis provides — not a price target but a framework for thinking. The structural window, two-cycle interaction, cost curve anchors, sodium-ion ceiling, Bayesian updating rules — these are analytical tools that remain useful regardless of which scenario materializes. A trader who internalizes this framework can update their view intelligently as monthly data arrives, rather than waiting for the next bank research note.

Chapter 5

Trading Implications — From Analysis to Action

5.1 What Makes Lithium Carbonate Analytically Distinct

Several structural characteristics differ meaningfully from mature commodity markets that traders are accustomed to. **Young futures market:** GFEX has existed less than three years; LME copper has traded since 1877. Price discovery is less efficient, basis relationships less stable, behavior under stress untested. **Bifurcated physical market:** Chinese-integrated supply chain (lower cost, dominant volume) versus Western-origin chain (higher cost, geopolitical premium). **Single geography dominance:** China's 48% consumption share creates tail risk that geographic diversification cannot hedge. **No mature recycling buffer:** demand shocks transmit directly to primary supply pressure. **Dynamic price ceiling from sodium-ion:** right tail of price distribution structurally truncated. **Two cycles at different speeds:** fast price cycle vs slow mining investment cycle, creating multi-year directional opportunities.

5.2 Physical Market Trading

Despite GFEX's growing role, the majority of global lithium carbonate volume changes hands through bilateral OTC contracts. Long-term offtake agreements (3–7 years) between major producers and battery makers remove volume from the freely tradeable market — reducing GFEX float for delivery. Spot transactions represent the marginal, price-setting volume.

The two-tier physical market and geopolitical premium trade: Buy Chinese-processed carbonate at GFEX spot price (lower, liquid). Source non-Chinese origin via Australian or South American processing (higher cost, less liquid). Sell Western-origin carbonate to Western automakers at premium price. The spread = Western-origin premium minus processing cost differential — and has been widening as IRA enforcement tightens.

Pilbara BMX as upstream price signal: BMX auction clearing prices reflect input cost for Chinese refiners. When BMX rises faster than GFEX carbonate, refining margins compress — early warning of production cuts. When BMX falls, margins expand — capacity restarts incentivized.

5.3 GFEX Futures Market — Mechanics and Trading Dynamics

Contract specifications: Ticker LC. Battery-grade lithium carbonate, minimum 99.5% Li_2CO_3 purity. 1-tonne lots. RMB per tonne. Daily price limits 10%. Trading margins 12% speculative, 11% hedging. Position limits for non-Chinese members 3,000 lots on most active contract.

Forward curve structure: Persistent contango during 2023–2024 reflected severe surplus. As physical tightening emerged through 2025–2026, the curve flattened toward backwardation. As of April 2026 with prices at 175,000 yuan/t following Zimbabwe ban, the curve has moved toward inverted structure consistent with deficit transition.

Calendar spread trade: If structural deficit widens through 2026–2028, deferred contracts (2027–2028) should rally toward spot. The spread compression trade — long deferred / short near-term — captures this transition without taking outright directional exposure.

The spodumene-to-carbonate spread (lithium's crack spread): Margin = GFEX carbonate price - (Spodumene BMX x 7.5 conversion) - processing costs. At April 2026: 175,000 - 53,108 - 30,000 ~ 92,000 yuan/t margin — highly profitable, incentivizing capacity restarts that will moderate the price spike. When this spread compresses toward zero, refiners cut output. When it expands sharply, restarts follow within

3–6 months.

5.4 Demand-Side Trading Signals

The CAAM-to-GFEX lead relationship: Monthly Chinese NEV sales typically lead GFEX price movements by 4–8 weeks. A trader monitoring CAAM data release dates (~10th of each month) and positioning on GFEX within 48 hours of strong readings can capture this informational advantage. Most GFEX participants focus on price data rather than upstream demand fundamentals — a systematic gap that fundamental analysis can exploit.

The inventory drawdown signal: 30,000-tonne threshold represents ~3 weeks of Chinese consumption. Below this level, physical buyers compete for spot material, driving backwardation within 2–4 weeks. Trader entering long position when inventory crosses 40,000 tonnes captures the move at lower entry cost than waiting for confirmed acute tightness.

5.5 Cross-Market Trading Opportunities

GFEX vs SGX basis represents premium Chinese buyers pay above international prices during domestic tightness — a real-time measure of China's demand intensity. **The carbonate-hydroxide spread** directly expresses the LFP-NMC chemistry battle: when LFP gains share, carbonate strengthens vs hydroxide. **Battery metals portfolio:** long lithium / short cobalt expresses LFP market share thesis with reduced sector exposure. Lithium-nickel relationship more complex through high-nickel NMC chemistry.

5.6 Risk Framework

Risk 1 — China policy shock (highest impact): CAAM monthly data is earliest signal — three consecutive months below 10% YoY triggers fundamental reassessment. Position management: any fundamental long should reduce by 30–50% if this triggers. **Risk 2 — Sodium-ion accelerating:** CATL quarterly disclosures break out sodium-ion volumes; quarter exceeding 10% triggers ceiling reassessment. **Risk 3 — SQM volumes-over-value:** lowest-cost producer continued production prevents self-correcting supply reduction. **Risk 4 — GFEX liquidity and position limits:** 3,000-lot daily limit constrains large positions; size conservatively. **Risk 5 — Basis risk:** physical and financial markets temporarily disconnect during stress.

5.7 Glencore's Specific Positioning

Glencore occupies an unusual position — simultaneously one of the world's largest physical commodity traders and a sophisticated financial market participant. This dual positioning creates specific advantages in lithium carbonate that pure financial or pure physical traders cannot easily replicate.

Physical market reach: Existing relationships with mining companies across Australia, South America, and Africa from coal, copper, cobalt, and nickel trading provide natural access to lithium upstream. **Western-origin premium opportunity:** Geographic reach across non-Chinese mining jurisdictions positions Glencore naturally to source, certify, and deliver Western-origin carbonate to premium-paying Western buyers as IRA and EU regulation deadlines tighten. **Singapore as trading hub:** Optimal geographic position — proximity to Chinese physical market participants, access to SGX international hedging, time zone overlap with Chinese GFEX hours and European/Australian business hours. **Cobalt and nickel experience as transferable expertise:** Frameworks used in cobalt trading translate directly to lithium carbonate. **The integrated physical-financial position:** Combining physical sourcing, transportation, storage, delivery, and financial hedging into unified positions captures value at multiple supply chain points simultaneously — the defining advantage of an integrated commodity trader in a market transitioning from physical to financialized.

5.8 Monitoring Framework

Weekly: GFEX inventory data (SMM/Mysteel) — <40K watch / <30K acute. GFEX forward curve shape — contango to flat = transition signal, backwardation = acute tightness. Pilbara BMX auction prices — >$1,200/t spodumene = restart incentives building, <$800/t = cuts coming.

Monthly: CAAM NEV sales (~10th of month) — >20% YoY bullish update, <10% YoY bearish update. GGII battery installation — LFP share >75% = carbonate accelerating. Chinese refinery operating rates — <70% = bullish supply tightening, >85% = supply response beginning.

Quarterly: CATL earnings — sodium-ion >10% of output = ceiling risk increasing. BYD production. Major miner production reports. USGS annual updates.

Event-driven: Mine closure/restart announcements, country export restrictions, CATL battery chemistry announcements, DLE commercial milestones, GFEX specification changes.

5.9 A Quantitative Researcher's Perspective

This section reflects the author's academic research in quantitative finance and international market integration, applied to the specific analytical challenges of the lithium carbonate market.

On price risk measurement in a young and non-normal market:

Standard VaR models — parametric, historical simulation, or Monte Carlo — assume relatively stable price change distributions. For lithium carbonate, this is deeply problematic. The 2021–2022 supercycle produced a 10-fold increase in 24 months — a move assigned essentially zero probability by any normal-distribution model calibrated to pre-2021 data. The distribution undergoes structural regime changes: the pre-GFEX market (pre-July 2023) was fundamentally different from the post-GFEX market — directly analogous to the structural break problem in international market integration research where the relationship between markets changes following liberalization events, making pre-event parameters unreliable for post-event risk estimation.

Regime-aware VaR framework: Identify current regime — surplus (2023–2025), transition (2025–2026, current), or deficit (2026–2028, projected). Apply regime-specific volatility estimates: surplus regime ~40–60% annualized, transition ~60–90%, deficit potentially >100% based on 2021–2022 experience. Do not use full historical lookback. Weight recent regime observations more heavily. Apply explicit stress scenarios from 2021–2022 episode regardless of statistical probability implied by current volatility estimates. This regime-aware approach — analogous to time-varying factor models in international market integration research — produces more reliable risk estimates than static historical VaR.

On market integration and GFEX maturation:

Research on international equity market integration provides a useful framework for anticipating GFEX evolution. Liberalization studies show integration increases rapidly in the initial period — new participants enter, arbitrage capital flows in, price discovery improves — then moderates as easy mispricings are eliminated and the market reaches equilibrium integration. GFEX is currently in the early phase: Chinese domestic prices are not yet fully integrated with international benchmarks. This creates persistent basis differentials, slower transmission of international demand signals, and arbitrage opportunities for participants accessing both markets. As GFEX progressively opens to international participants — gradually, through January 2026 designation as internationally accessible instrument — integration will increase. The integration-phase opportunities are temporary but potentially multi-year.

On the Tail Fragility Index and battery metals systemic risk:

The Tail Fragility Index developed in the author's doctoral research — measuring tendency of markets to experience simultaneous extreme losses under stress — has potential application to battery metals portfolio construction not explored in existing literature. Battery metals (lithium, cobalt, nickel, manganese, graphite) are all components of the same downstream value chain (EV batteries) and subject to the same macro demand driver. A demand shock to EV markets produces simultaneous price stress across all battery metals; a Chinese policy shock transmits instantly through the entire supply chain.

A portfolio appearing diversified under normal conditions may be highly concentrated in tail scenarios — the opposite of what traditional correlation diversification suggests. This is directly analogous to the finding in international market integration research that cross-market correlations increase dramatically during financial crises — when diversification is most needed, it disappears. Standard correlation-based position sizing systematically underestimates tail risk of a battery metals portfolio. **A TFI-based risk overlay — monitoring co-crash tendency of battery metals under stress — would provide a more reliable risk management tool than VaR models that treat cross-metal correlations as stable parameters.**

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End of Report